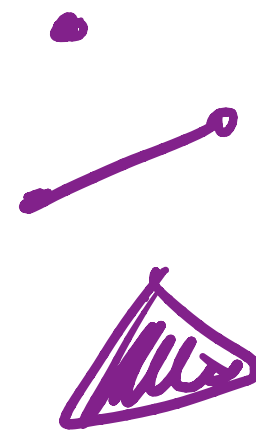


# Here Comes the Homology

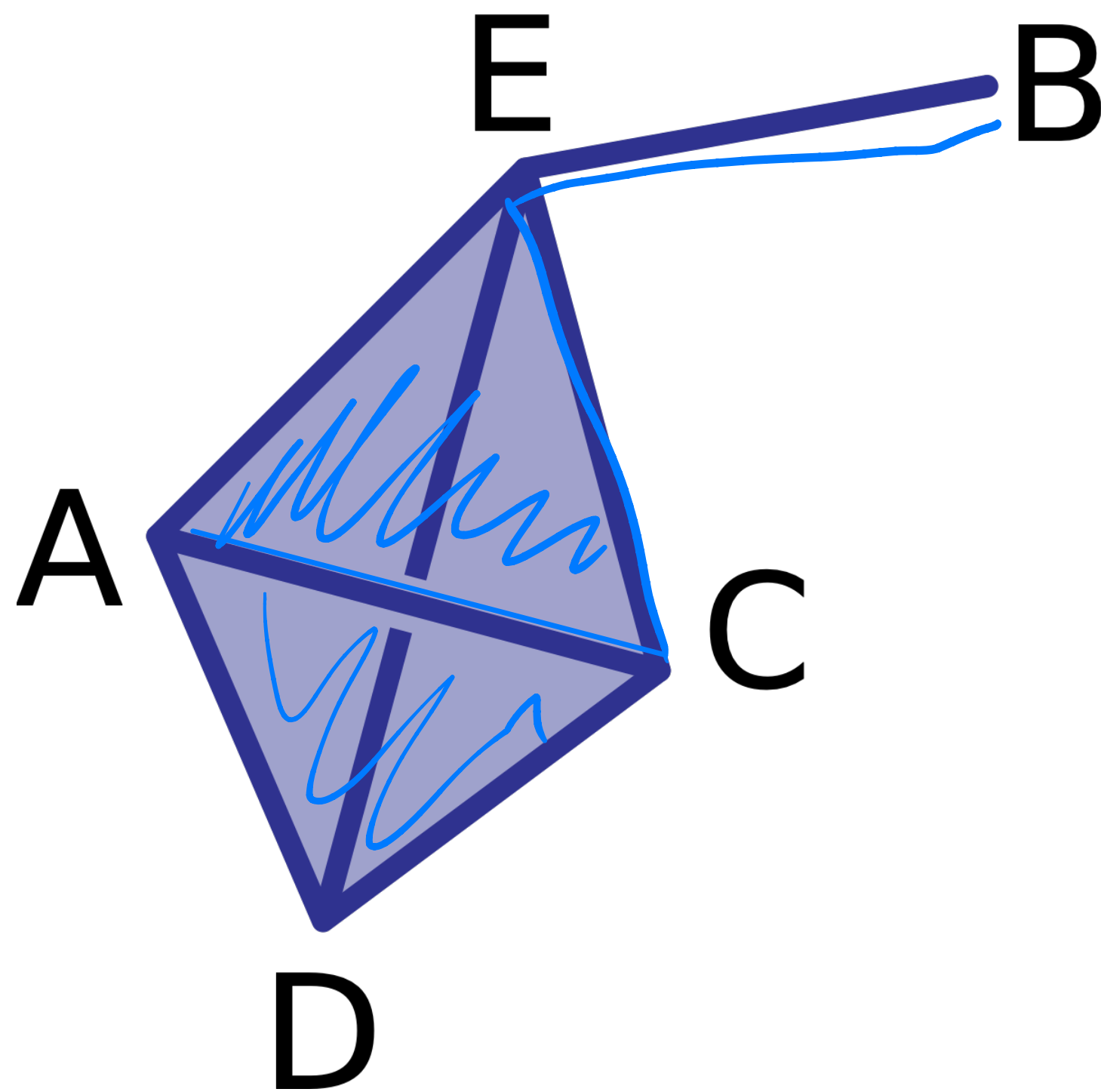
Will we get there? — Yes!

# $p$ -Chains



Let  $K$  be a simplicial complex and fix a dimension  $p$ .

A  $p$ -chain is a finite formal sum of  $p$ -simplices in  $K$ , written  $\alpha = \sum_i a_i \sigma_i$



# Chain Group

The collection of  $p$ -chains with addition is called the  $p$ th-chain group (vector space),  $C_p(K)$ .

Closure:  $c + d \in C_p(K)$

Associative:  $(c + d) + e = c + (d + e)$

Commutative:  $c + d = d + c$

Zero element:  $c + 0 = 0 + c = c$

Inverses:  $c + (-c) = 0$

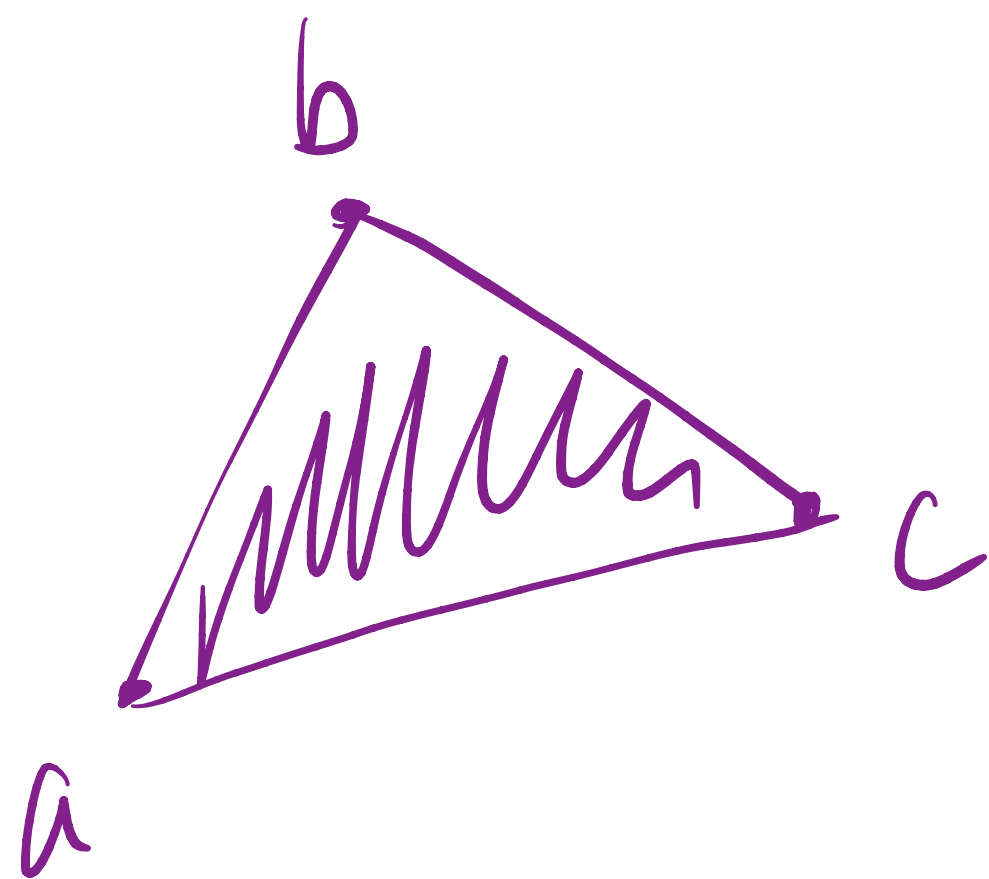
# Boundary Maps

$$\partial_p : C_p(K) \longrightarrow C_{p-1}(K)$$

$$\sigma = [v_0, \dots, v_p] \longmapsto \sum_{j=0}^p [v_0, \dots, \widehat{v_j}, \dots, v_p]$$

$$\underline{\mathbb{Z}_2}$$

$$2 \equiv 0 \pmod{2}$$



$$\partial_2 : C_2(K) \longrightarrow C_1(K) \longrightarrow C_0(K)$$

$$\sigma : abc$$

$$ab + ac + bc$$

$$\partial_{p+1} \partial_p(K) = 0$$

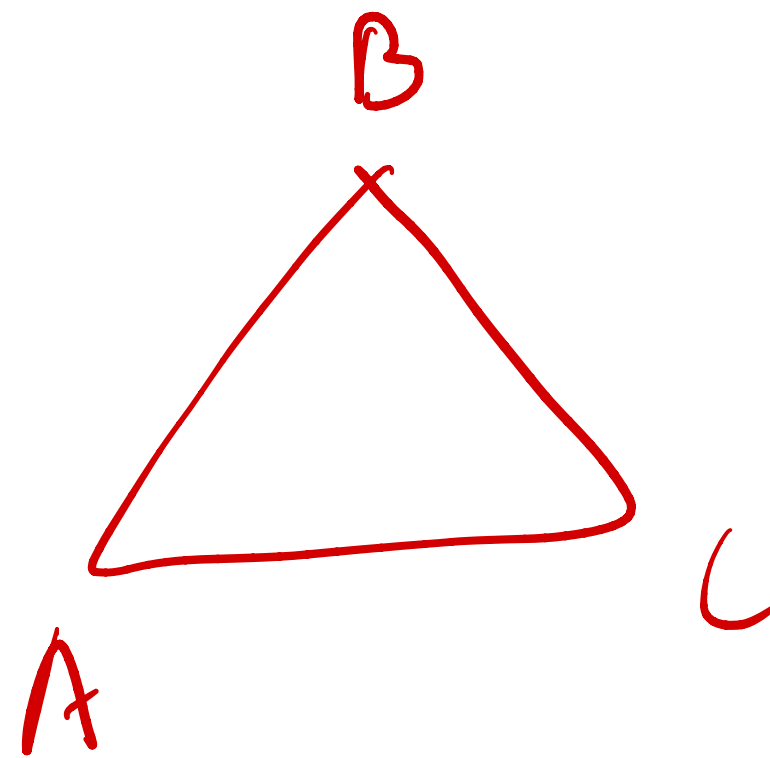
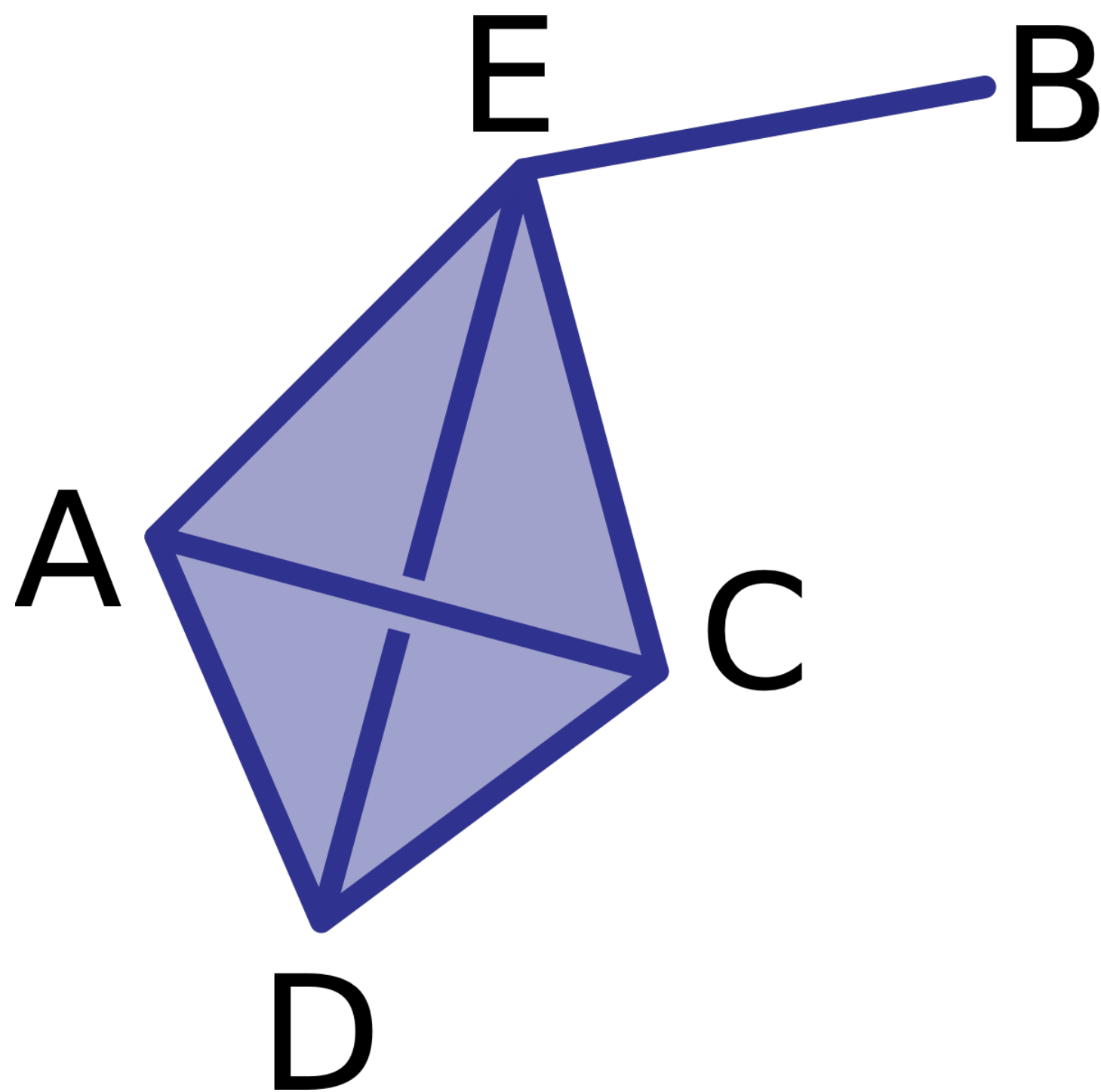
$$\begin{aligned} & \hat{a}b + \hat{a}b \\ & + \hat{a}\hat{c} + \hat{a}c \\ & + \hat{b}\hat{c} + \hat{b}c \end{aligned}$$

$$= \cancel{a} + \cancel{b} + \cancel{a} + \cancel{c} + \cancel{b} + \cancel{c}$$

# Cycles

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the kernel of  $\partial_p$  is called a  $p$ -cycle.  $Z_p(K) = \ker(\partial_p) = \{c \in C_p(K) \mid \partial_p c = 0\}$



$$\begin{array}{ccc} C_2 & \xrightarrow{\partial_2} & C_1 & \xrightarrow{\partial_1} & C_0 \\ ABC & & AB+BC & & 0 \\ & & +AC & & \\ & & \hline & & \alpha \end{array}$$

$$\alpha \in \ker(\partial_1)$$

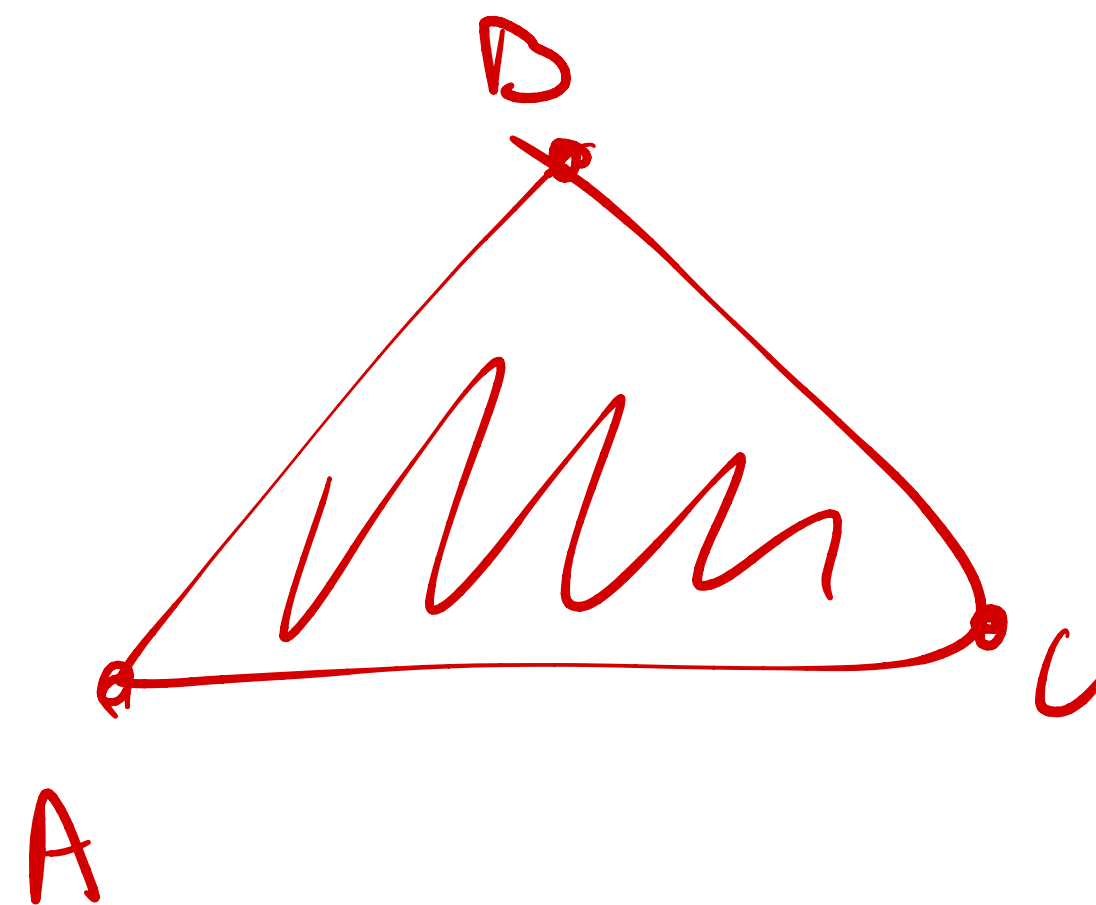
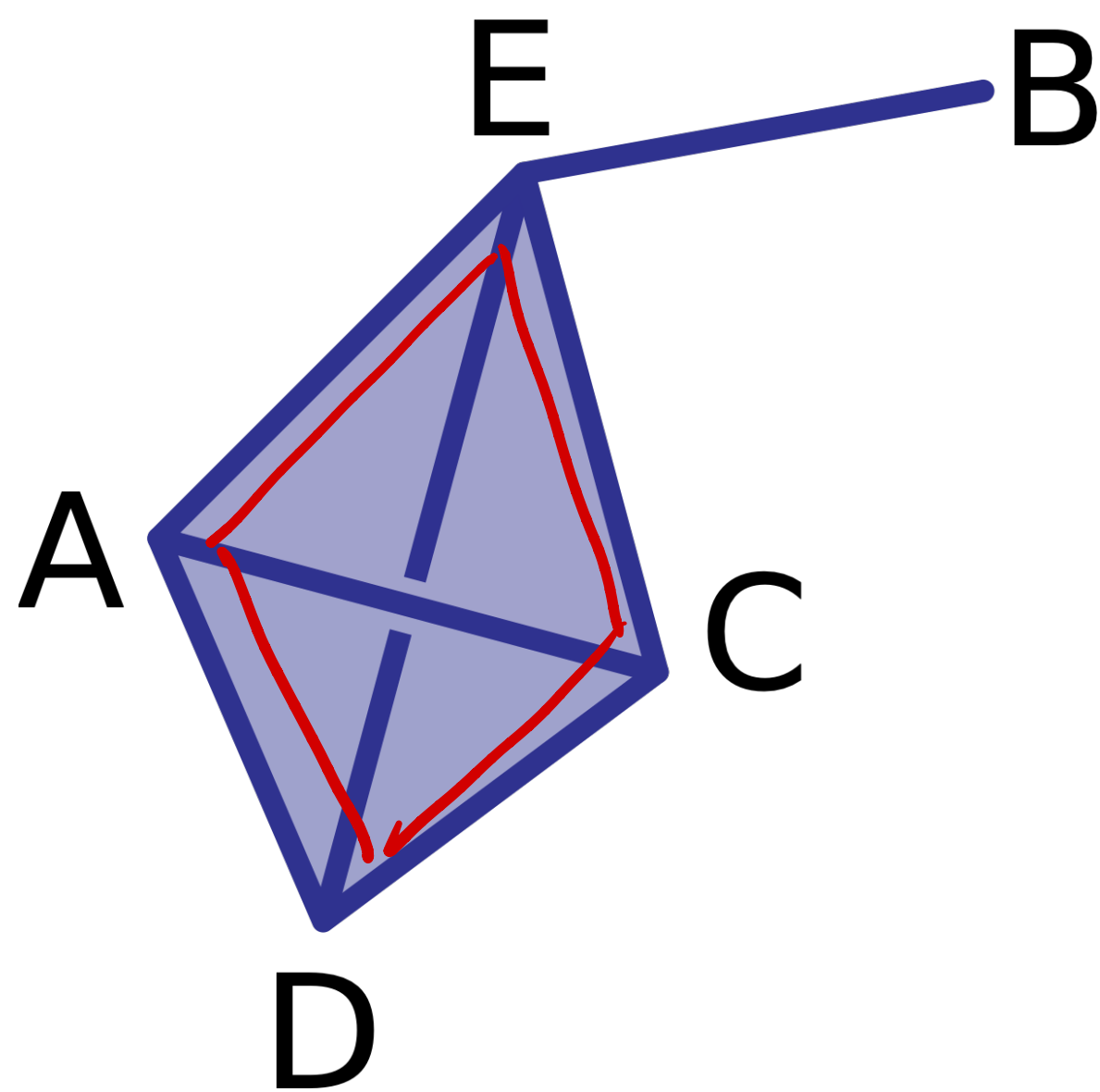
The collection of  $p$ -cycles forms a subspace  $Z_p(K) \subseteq C_p(K)$

# Boundaries

$$C_{p+1}(K) \xrightarrow{\partial_{p+1}} C_p(K) \xrightarrow{\partial_p} C_{p-1}(K)$$

A chain in the image of  $\partial_{p+1}$  is called a  $p$ -boundary

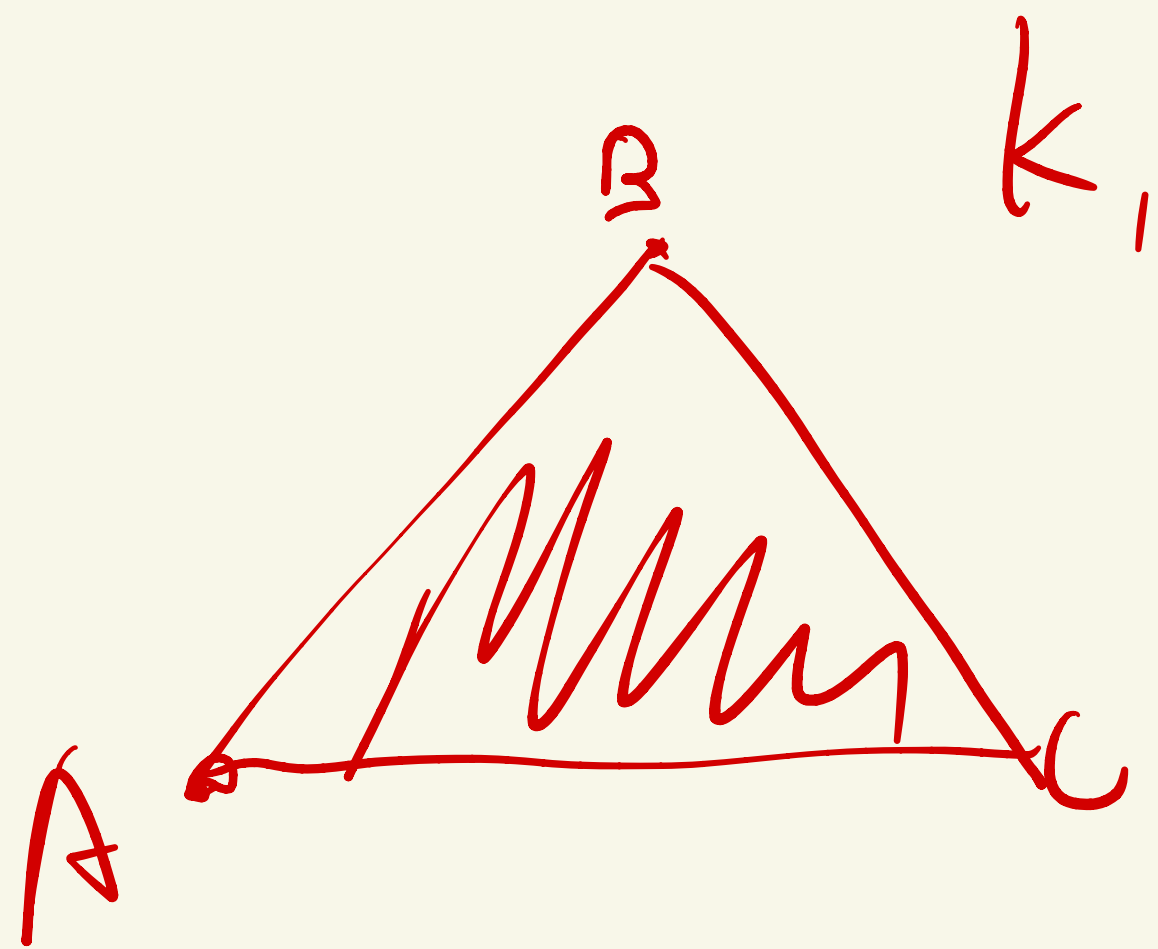
$$B_p(K) = \text{im}(\partial_{p+1}) = \{\partial_{p+1}c \mid c \in C_{p+1}(K)\}$$



$$\begin{array}{l} \textcircled{C_2} \xrightarrow{\partial_2} C_1 \\ \text{ABC} \quad \left. \begin{array}{l} AB + BC \\ + AC \end{array} \right\} \\ \qquad \qquad \qquad \underbrace{\hspace{1cm}} \\ \qquad \qquad \qquad \in B_1(K). \end{array}$$

$$B_p(K) \subseteq Z_p(K) \subseteq C_p(K).$$

The collection of  $p$ -boundaries forms a subspace  $B_p(K) \subseteq C_p(K)$



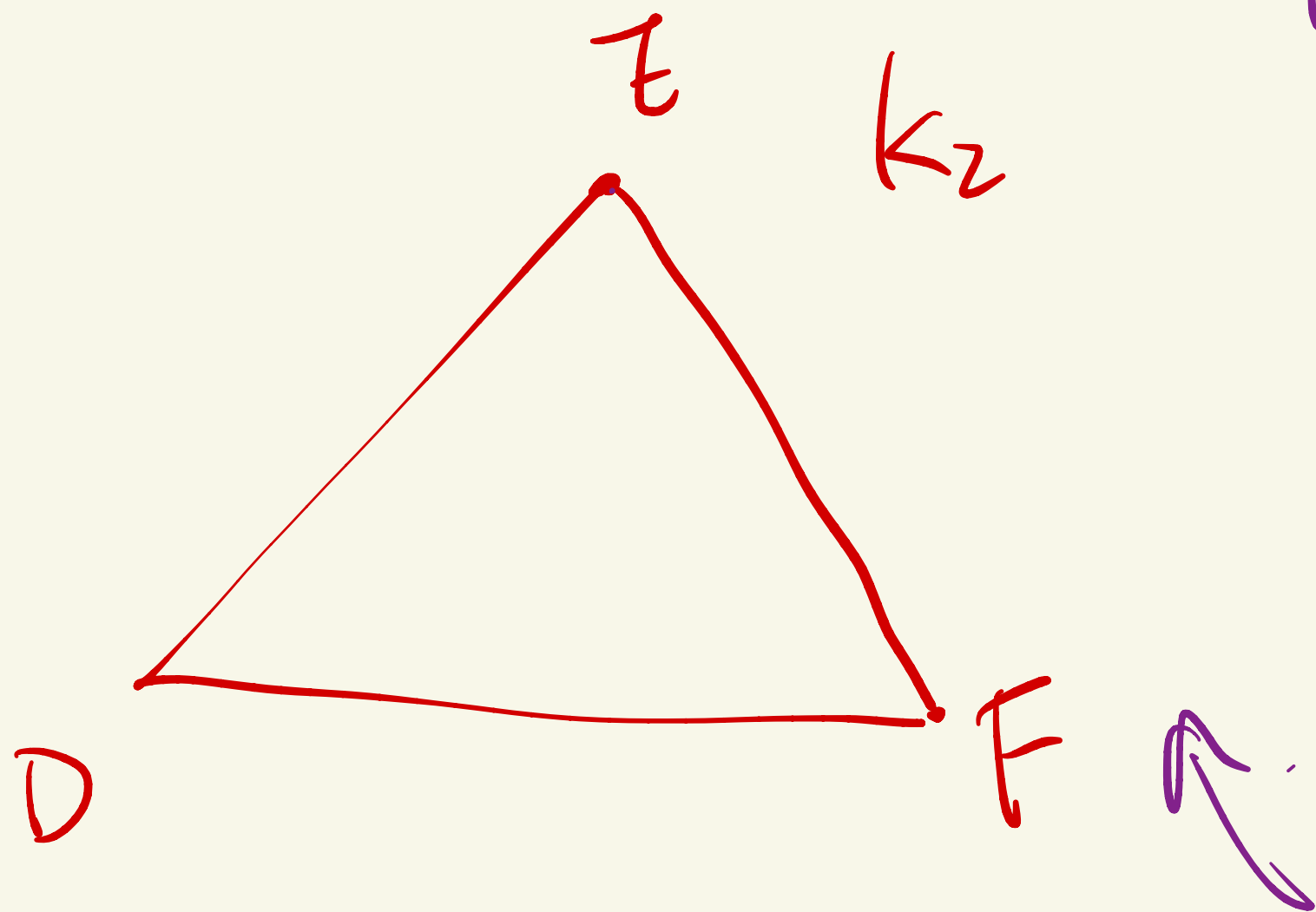
$$C_2 \xrightarrow{\alpha_2} C_1 \xrightarrow{\alpha_1} C_0$$

$$K_1: ABL \xrightarrow{\alpha_2} \begin{pmatrix} AC+BC \\ +AB \end{pmatrix}$$

$$\begin{pmatrix} DE+DF \\ +EF \end{pmatrix}$$

$$K_2 \quad C_2 = 0$$

$$\text{Im}(\alpha_2)$$

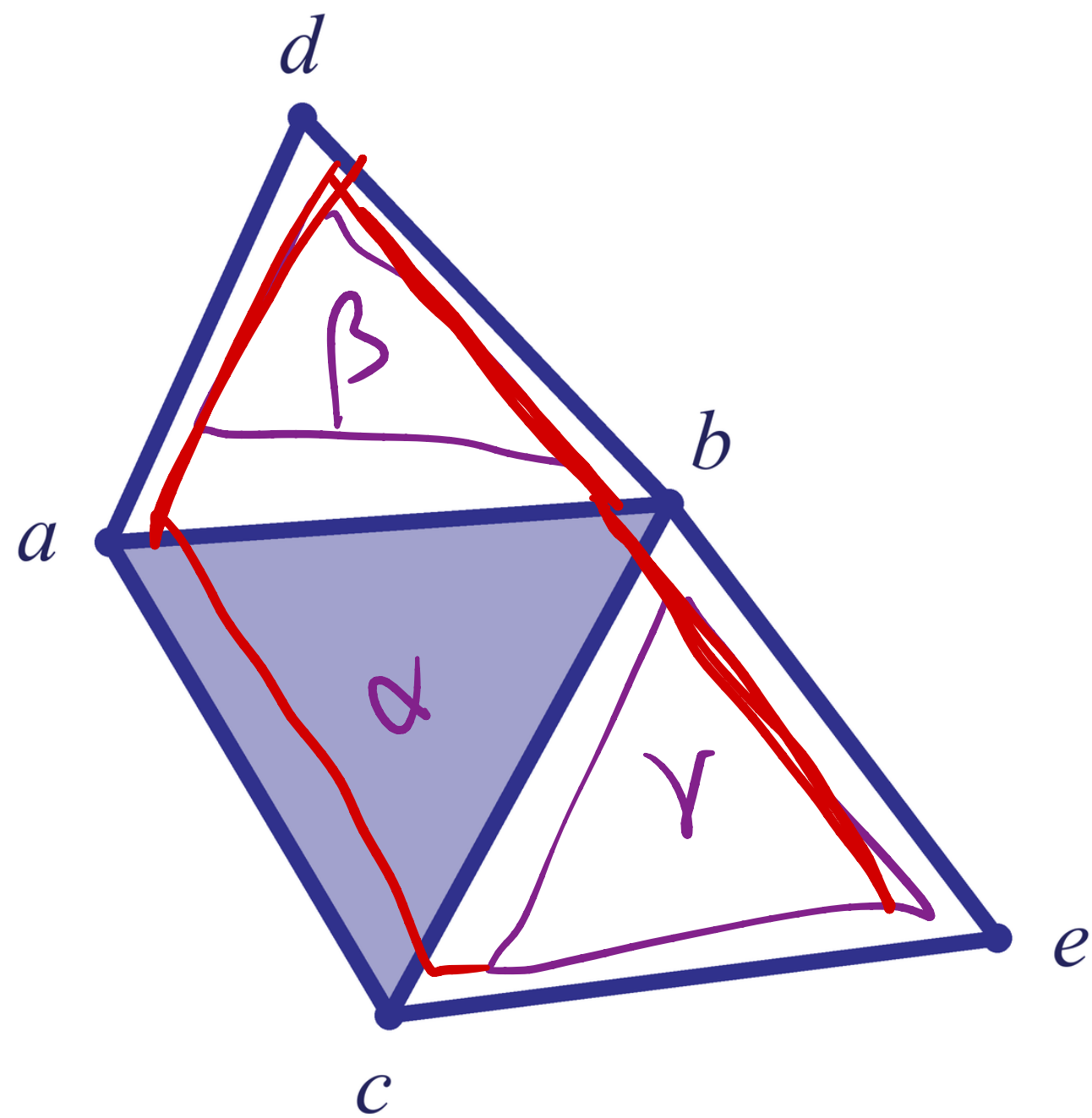


$$K_2: \{D, E, F, DE, DF, \underline{EF}\}.$$

$C_1$

$$B_1(K) \subseteq Z_1(K)$$

# Example: Cycles and Boundaries



What are the generators of  $B_1(K)$ ? Of  $Z_1(K)$ ?

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0.$$

abc.

$$\left. \begin{array}{l} \alpha: ab+bc+ac \\ \beta: ad+ab+db \\ \gamma: cb+ce+be \end{array} \right\} \begin{array}{l} 0 \\ 0 \\ 0 \end{array}$$

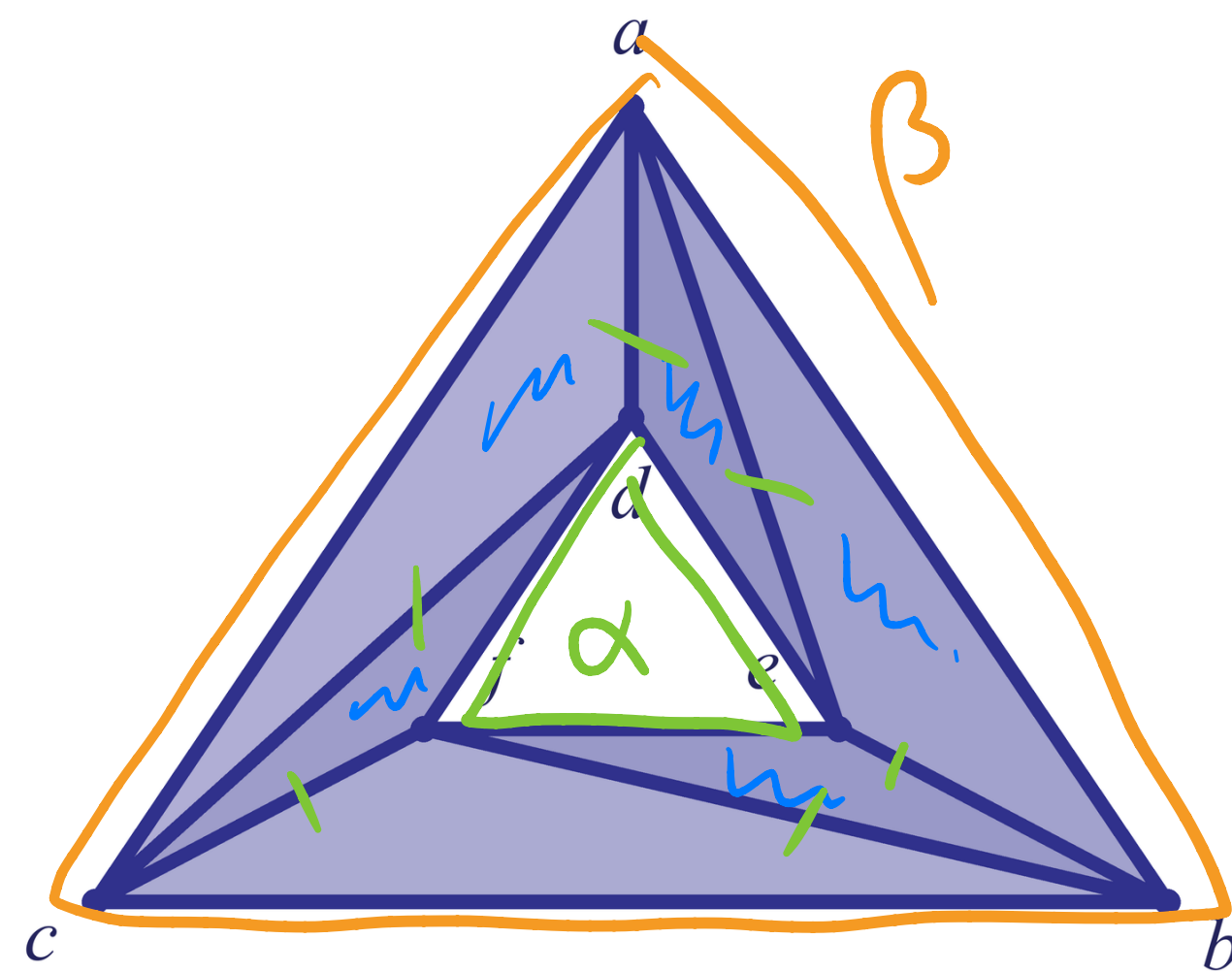
$$B_1(K) = \langle \underline{\alpha} \rangle.$$

$$\hookrightarrow Z_1(K) = \langle \alpha, \beta, \gamma \rangle.$$

$$ad+db+be+ec+ca = \alpha + \beta + \gamma$$

$$\begin{aligned} &= \{0, \alpha, \beta, \gamma, \\ &\quad \alpha+\beta, \alpha+\gamma, \\ &\quad \beta+\gamma, \\ &\quad \alpha+\beta+\gamma\}. \end{aligned}$$

# Example: Cycles and Boundaries



$$\begin{aligned} fd + de + fe &= \alpha \\ ab + bc + ac &= \beta \end{aligned}$$

What are the generators of  $B_1(K)$ ? Of  $Z_1(K)$ ?

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0$$

$C_2$   
 $acd$   
 $cdf$   
 $ade$   
 $abe$   
 $bef$   
 $bcf$

$C_1$   
 $\gamma_1: ac + cd + ad$   
 $\gamma_2: cd + df + cf$   
 $\gamma_3: ad + de + ae$   
 $\gamma_4: ab + be + ae$   
 $\gamma_5: be + ef + bf$   
 $\gamma_6: bc + cf + bf$

$$\gamma_1 + \dots + \gamma_6 = \alpha + \beta$$

$$\gamma_1 + \dots + \gamma_6 + \alpha = \beta$$

$$B_1(K) = \langle \gamma_1, \dots, \gamma_6 \rangle$$

$$Z_1(K) = \langle \gamma_1, \dots, \gamma_6, \alpha \rangle$$

$$\begin{aligned} &\gamma_1 + \dots + \gamma_6 + \alpha \\ &= ab + bc + ac + \cancel{fd} + \cancel{de} + \cancel{fe} \\ &\quad + \cancel{fd} + \cancel{de} + \cancel{fe} \\ &= ab + bc + ac \\ &= \beta \end{aligned}$$

Homology for Real Now

# Equivalence Classes

An equivalence relation  $\sim$  on a set  $S$  satisfies:

- Reflexivity:  $x \sim x$
- Symmetry: if  $x \sim y$ , then  $y \sim x$
- Transitivity: if  $x \sim y$  and  $y \sim z$ , then  $x \sim z$

The equivalence class of  $x$  consists of all elements equivalent to  $x$ :

$$[x] = \{y \in S \mid y \sim x\}$$

$\uparrow$  a set of elements, not just an element.

Ex. On  $\mathbb{Z}$ , define  $m \sim n$  when  $m - n$  is even.

$$\begin{aligned}
 & \uparrow \quad \downarrow \quad \downarrow \\
 & [0], [1] = [3] = [13] \dots \\
 & \uparrow \\
 & = [2] \\
 & = [52] \dots
 \end{aligned}$$

$$2 - 6 = -4 \quad 2 \sim 6$$

$$3 - 11 = -8 \quad 3 \sim 11$$

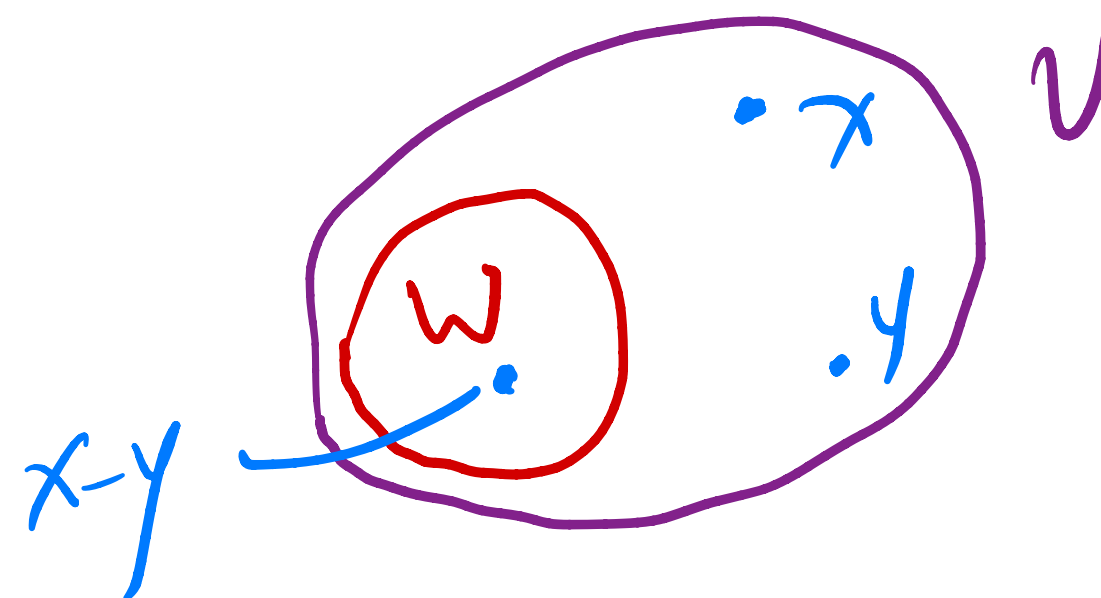
$$4 - 7 = -3 \quad 4 \not\sim 7$$

$$0 - 12 = -12 \quad 0 \sim 12$$

$$0 - 2n = -2n \text{ even} \Rightarrow 0 \sim 2n \quad \forall n.$$

$$1 - (2n+1) = -2n \text{ even} \Rightarrow 1 \sim (2n+1) \quad \forall n.$$

# Quotient Space



Let  $V$  be a vector space over a field  $k$ . Let  $W \subset V$  be a subspace.

Define  $\sim$  on  $V$  by  $x \sim y$  iff  $x - y \in W$ . The equivalence class of  $x$  is denoted



$$[x] = x + W = \{ \underline{x} + \underline{w} : w \in \underline{W} \}$$

The quotient space  $\underline{V}/\underline{W}$  is then defined as  $\{ [x] \mid x \in V \}$ . This is also a vector space with:

• Scalar multiplication:

$$a[x] = [ax]$$

• Addition:

$$[x] + [y] = [x + y]$$

equivalence classes.  $\mathbb{R}$

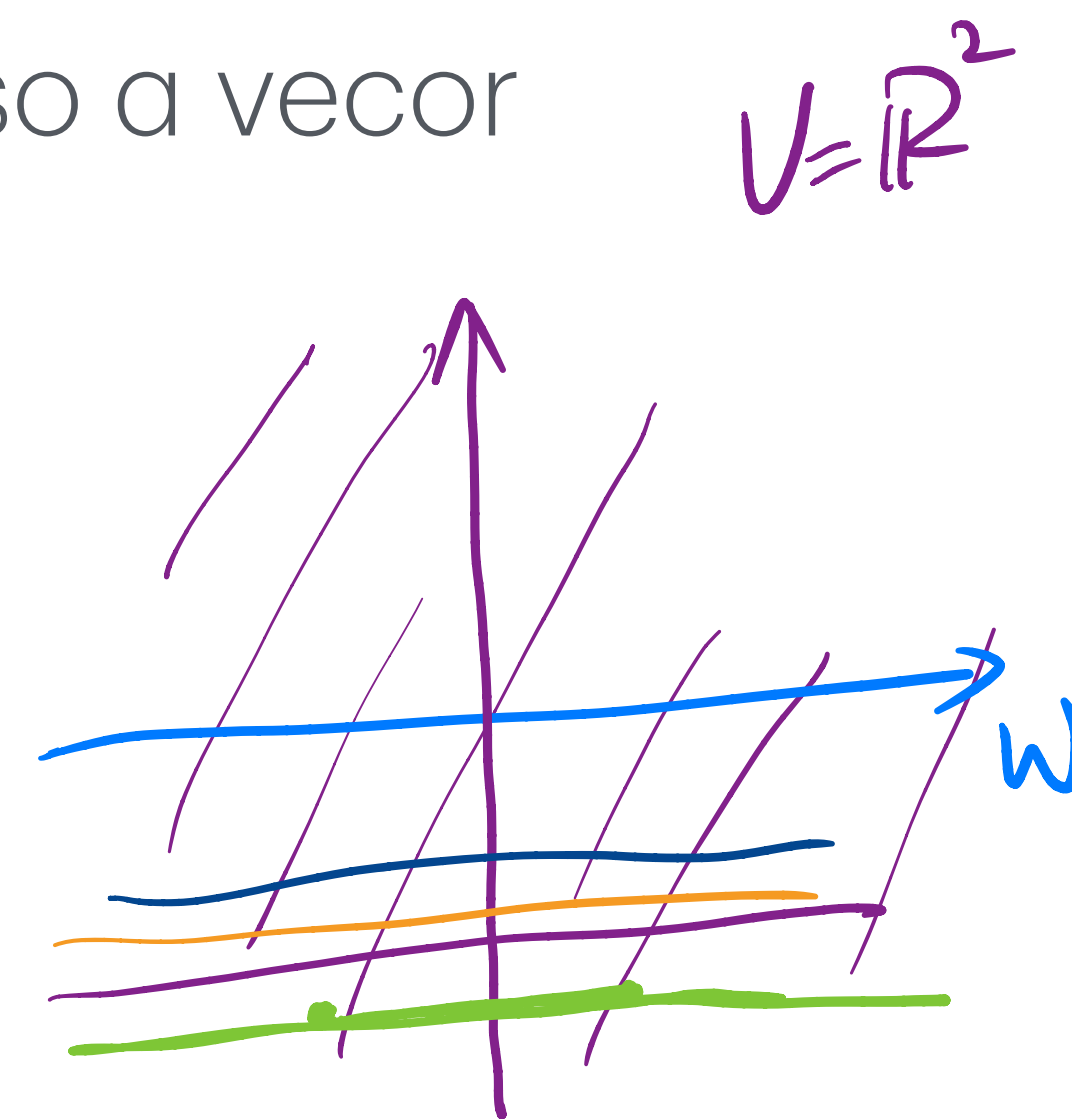
$$V = \mathbb{R}^2, W: x\text{-axis. } W = \text{span}\{(1, 0)\}$$

$$(x_1, y_1) \sim (x_2, y_2) \text{ if } (x_1, y_1) - (x_2, y_2) \in W.$$

$$\text{need } y_1 - y_2 = 0.$$

$$y_1 = y_2.$$

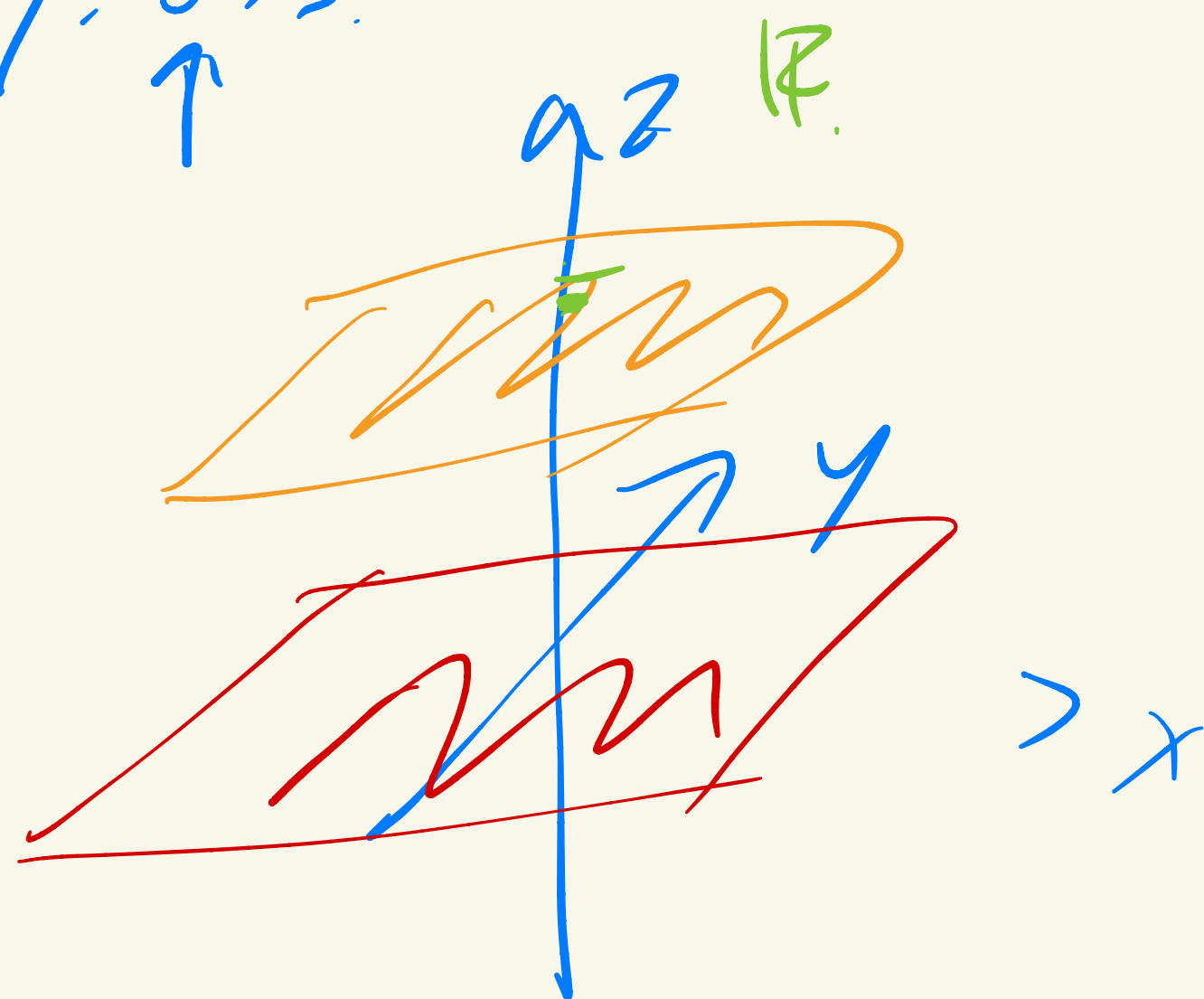
$$V/W = \{ [x] \}$$



$$V = \mathbb{R}^3$$

$W = xy\text{-plane}$ .

$$\langle (x, y, 0) \rangle$$



$$(x_1, y_1, z_1) \sim (x_2, y_2, z_2)$$

$$\nexists (x_1, y_1, \underline{z_1}) - (x_2, y_2, \underline{z_2}) \in W.$$

$$\underline{z_1 - z_2 = 0.}$$

$V/W$ .  $\mathbb{R}^3/W$ . identified with  $\mathbb{R}$

$$\dim(\mathbb{R}^3) = 3$$

$$\dim(W) = 2.$$

$$\dim(\mathbb{R}^3/W) = 1$$

$$\dim(V/W) + \dim(W) = \dim(V).$$

# Homology!!

$$\begin{aligned} [\alpha] &= \{ \alpha + w \mid w \in B_1(K) \} \\ &= \{ \underset{\uparrow \alpha}{\alpha + 0}, \underset{\uparrow 0}{\alpha + \alpha} \} = [0]. \end{aligned}$$

The  $p$ th homology group is the quotient space

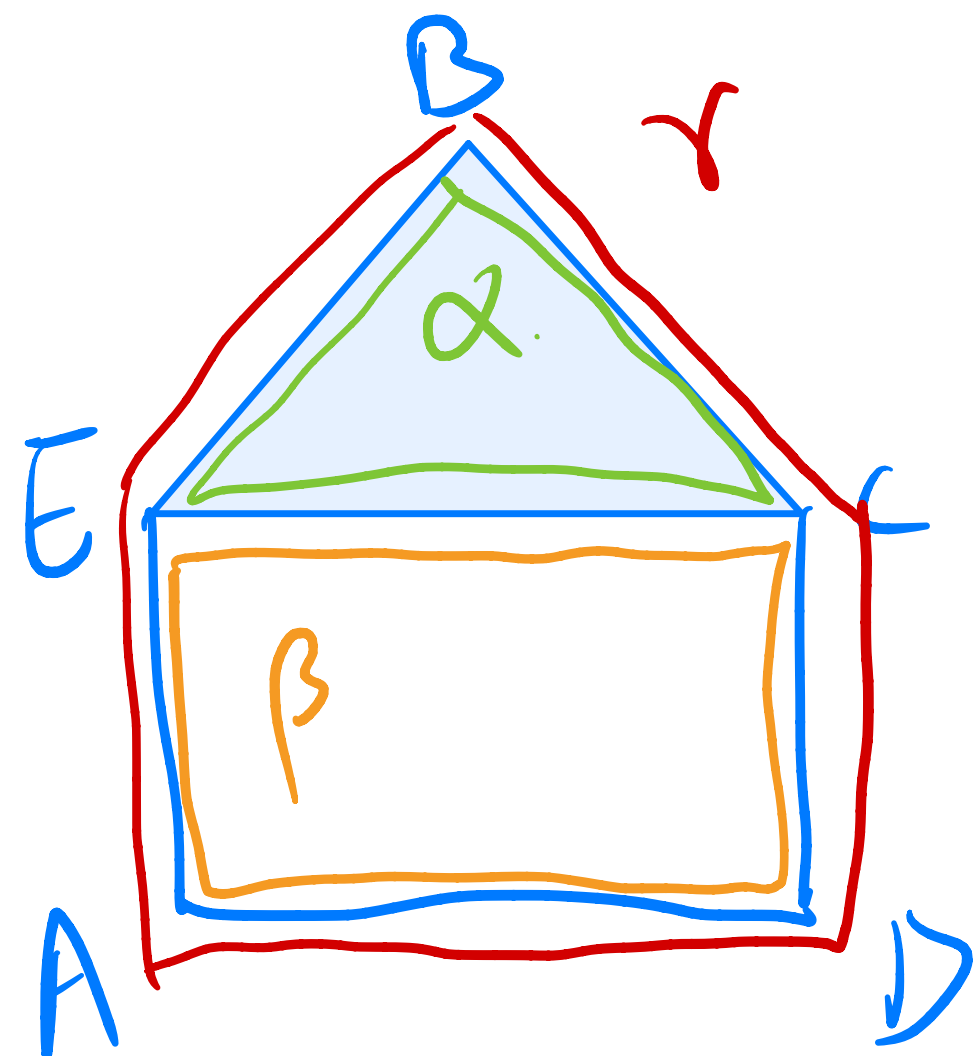
$$H_p(K) := \underset{\substack{\text{p-cycles} \\ \nwarrow}}{Z_p(K)} / \underset{\substack{\text{p-boundary} \\ \swarrow}}{B_p(K)}$$

two cycles are considered equivalent if they differ by a boundary.

$$x \sim y \iff x - y \in B_p(K)$$

$$B_1(K) = \langle \alpha \rangle$$

$$\cap_1 Z_1(K) = \langle \alpha, \beta \rangle$$



$$G \xrightarrow{\partial_2} \underset{\alpha}{C_1} \longrightarrow G.$$

$BCE.$

$$V = Z_1(K) = \{ 0, \alpha, \beta, \alpha + \beta \}.$$

$$W = B_1(K) = \{ 0, \alpha \}.$$

$$\begin{aligned} H_1(K) &= Z_1(K) / B_1(K) = \{ [0], [\beta] \} \\ &= \langle [\beta] \rangle \end{aligned}$$

$$\dim(H_1) = 1$$

$$\begin{aligned} [\beta] &= \{ \beta + w \mid w \in B_1(K) \} \\ &= \{ \beta + 0, \beta + \alpha \} \\ &= \{ \beta, \gamma \} \\ &= [\gamma]. \end{aligned}$$

Example: What is  $H_1(K)$ ?

$$C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0.$$

$ABC \quad AB+BC+CA.$

$$\partial_2(ABC) = \alpha.$$

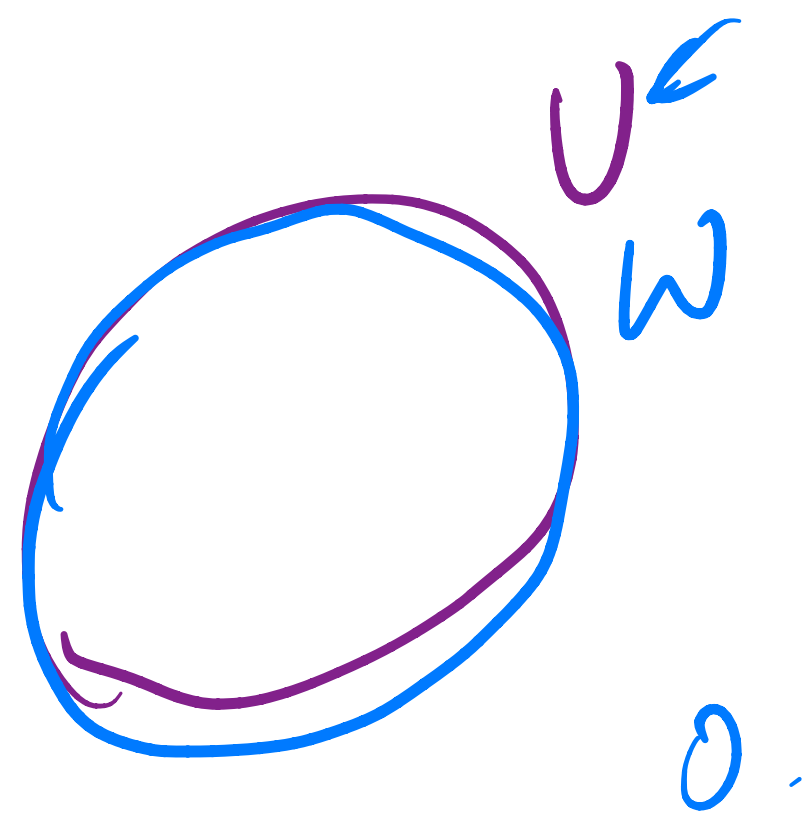
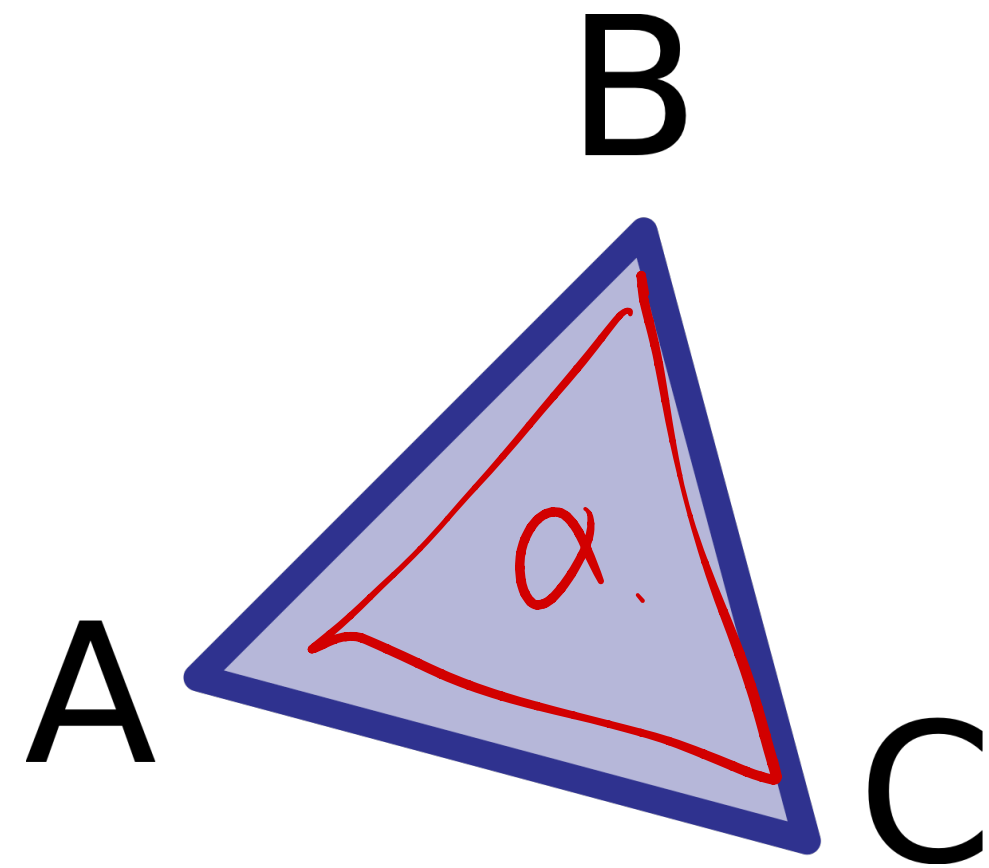
$$B_1(K) = \langle \alpha \rangle.$$

$$\partial_1(\alpha) = 0.$$

$$Z_1(K) = \langle \alpha \rangle.$$

$$B_1(K) = Z_1(K). \leftarrow$$

$$H_1(K) = Z_1(K) / B_1(K) = 0.$$



$$Z_1(K) = \{0, \alpha\}.$$

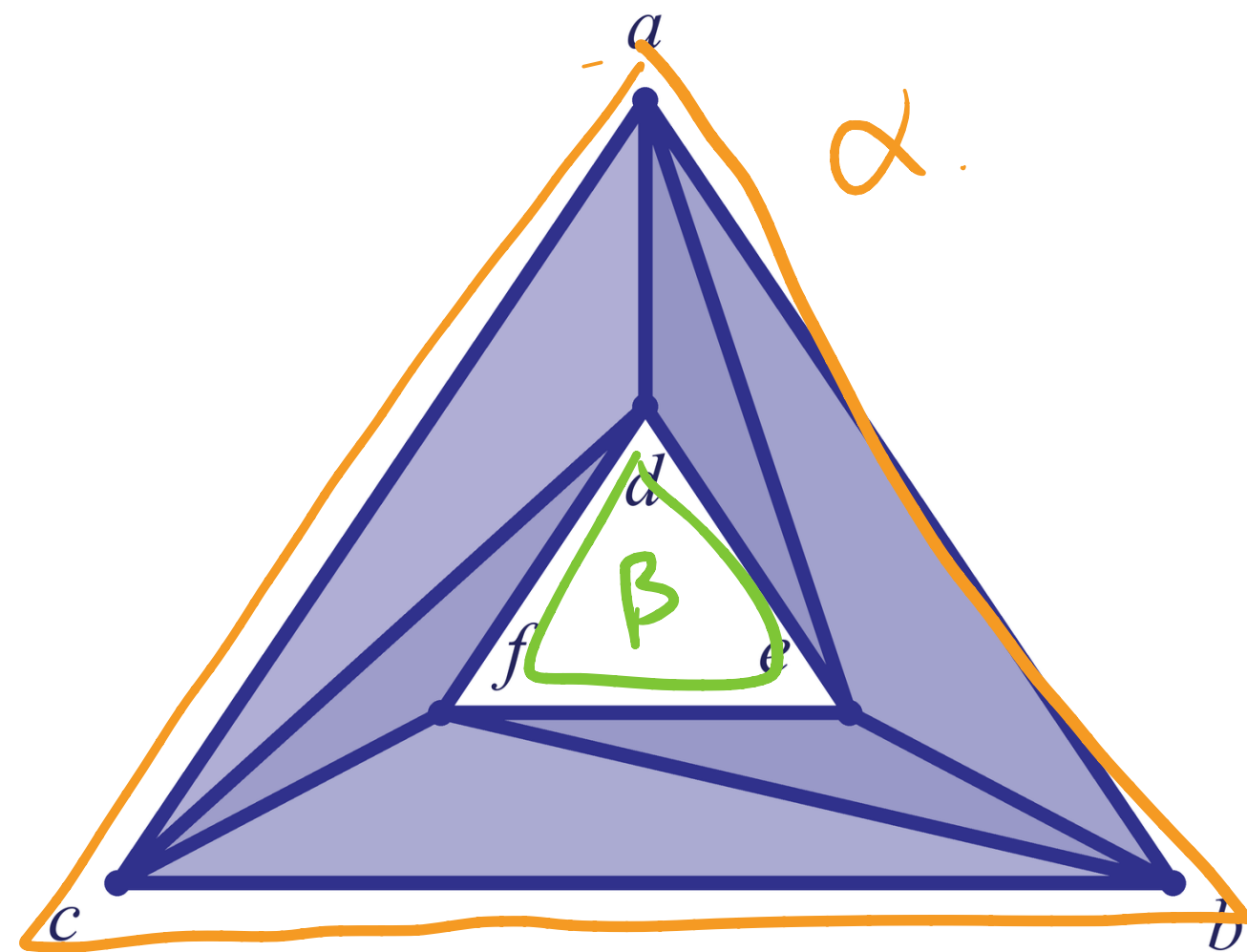
$$[\alpha] = \{\alpha + 0, \alpha + \alpha\}.$$

$$= \{\alpha, 0\}.$$

$$= [0].$$

Example: What is  $H_1(K)$ ?

$$\dim(H_1) = \dim(Z_1) - \dim(B_1)$$



$$B_1(K) = \langle \gamma_1, \dots, \gamma_6 \rangle$$

$$Z_1(K) = \langle \gamma_1, \dots, \gamma_6, \alpha \rangle$$

$$\dim(Z_1(K)) = 7$$

$$\dim(B_1(K)) = 6$$

$$\dim(H_1(K)) = 1$$

$C_2$   
acd  
cdf  
ade  
abe  
bef  
bcf

$$\partial_2: C_2 \rightarrow C_1 \rightarrow C_0$$

$$\gamma_1: ac + cd + ad$$

$$\gamma_2: cd + df + cf$$

$$\gamma_3: ad + de + ae$$

$$\gamma_4: ab + be + ae$$

$$\gamma_5: be + ef + bf$$

$$\gamma_6: bc + cf + bf$$

$$H_1(K) = \langle [\alpha] \rangle$$

$$\alpha \sim \beta$$

$$\alpha = \beta + \underbrace{\gamma_1 + \dots + \gamma_6}_{\in B_1(K)}$$

$$\Rightarrow [\alpha] = [\beta] \neq [0]$$

homotopy equivalent  $\Rightarrow$  same homology.

diff. homology  $\Rightarrow$  not homotopy equivalent.

"disc"



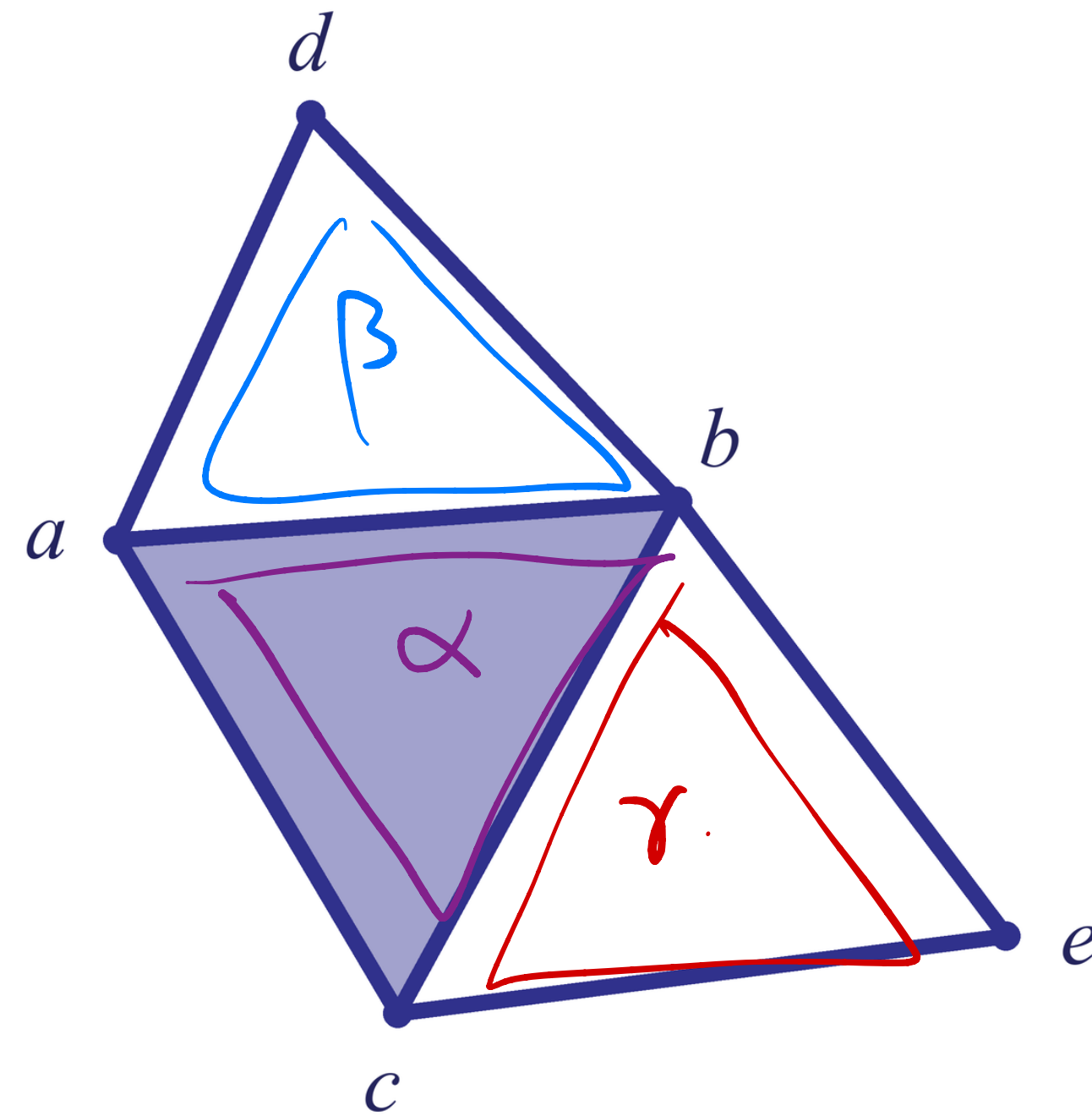
$$H_1(K_1) \neq H_1(K_2)$$

$$K_1 \not\sim K_2.$$

"annulus"



Example: What is  $H_1(K)$ ?



$$Z_1(K) = \langle \alpha, \beta, \gamma \rangle$$

$$= \{0, \alpha, \beta, \gamma, \alpha+\beta, \alpha+\gamma, \beta+\gamma, \alpha+\beta+\gamma\}$$

$$B_1(K) = \langle \alpha \rangle = \{0, \alpha\}$$

$$H_1(K) = Z_1(K) / B_1(K)$$

$$[0] = \{0, \alpha\} = [\alpha]$$

$$[\beta] = [\beta, \beta+\alpha]$$

$$[\gamma] = [\gamma, \gamma+\alpha]$$

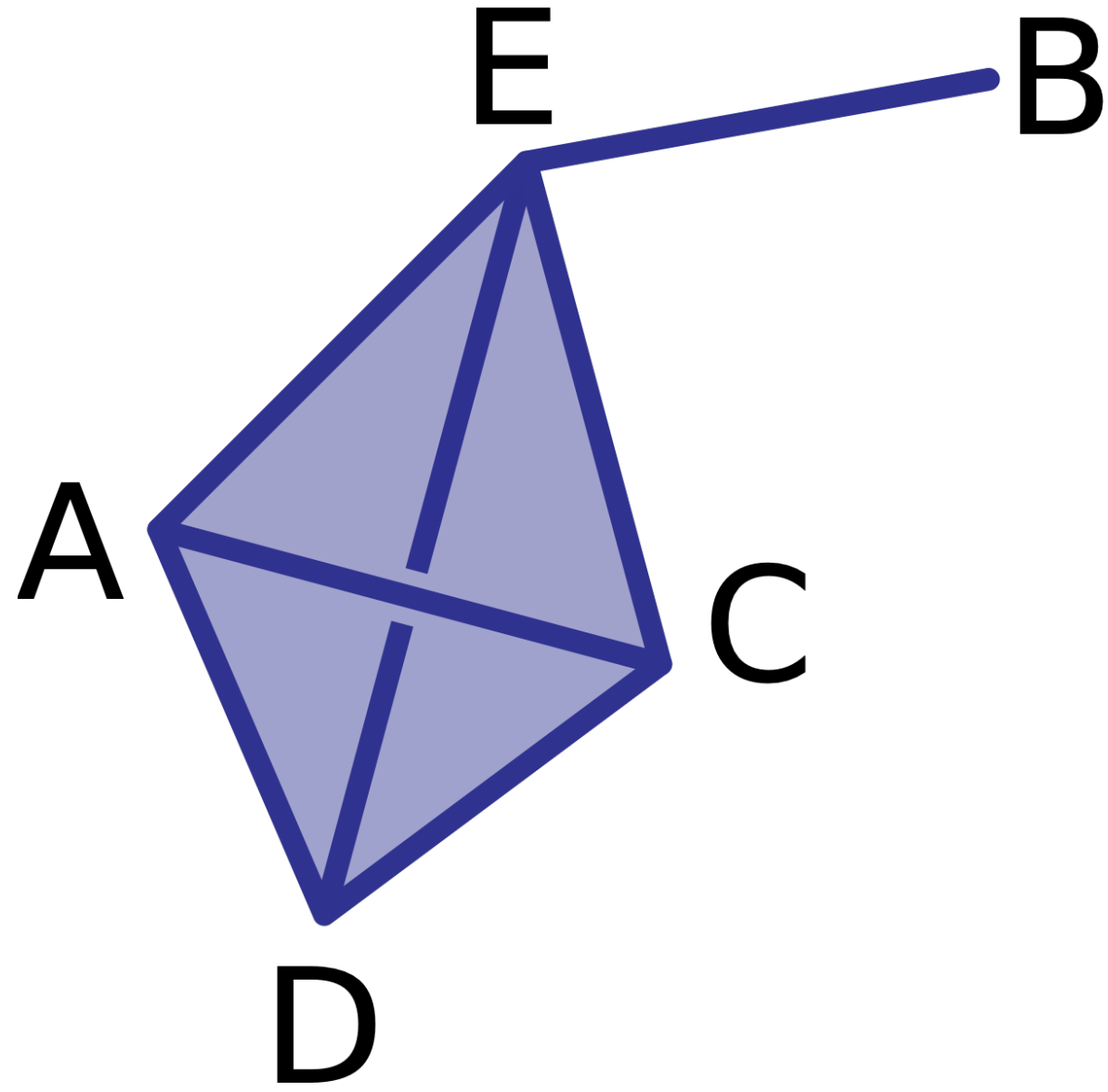
$$[\beta+\gamma] = [\beta+\gamma, \beta+\gamma+\alpha]$$

$$[\beta] + [\gamma] = [\beta+\gamma]$$

$$H_1(K) = \{[0], [\beta], [\gamma], [\beta+\gamma]\}$$

$$= \langle [\beta], [\gamma] \rangle$$

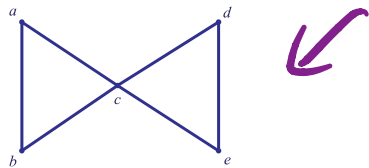
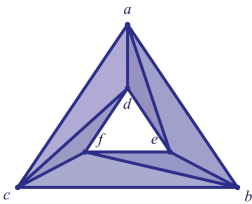
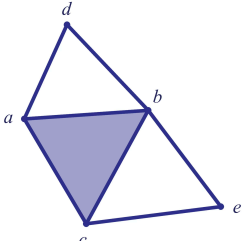
Example: What is  $H_2(K)$ ?



# Betti Numbers

$\beta_k = \dim H_k = \dim Z_k - \dim B_k$

- $\beta_0$ : number of connected components ↙
- $\beta_1$ : number of independent loop classes ↙
- $\beta_2$ : number of independent 2-dimensional homology classes ↙

|                                                                                     | $\beta_0$ | $\beta_1$ | $\beta_2$ |
|-------------------------------------------------------------------------------------|-----------|-----------|-----------|
|  | 1         | 2         | 0         |
|  | 1         | 1         | 0         |
|  | 1         | 2         | 0         |

$H_2 = 0$

# Take away

- We want to compute the “shape”: we turn “shape” into linear algebra; homology is the machinery
- $p$ -simplex: basis vector
- $p$ -chains: vectors
- Boundary map: matrix
- Cycles: kernel of a matrix
- Boundaries: image of a matrix
- Homology: cycles modulo trivial cycles

# Homework Presentation

- Let  $K$  be the simplicial complex of a hollow tetrahedron (no interior). Write a basis for  $C_2$ ; compute  $B_2$ ,  $Z_2$ , and  $H_2$ . Now add back the interior; compute  $H_2$  (we know  $B_2$  and  $Z_2$  from last week)
- Consider a complex  $K = \{A, B, C, D, E, AB, BC, AC, CD, AD, CE, DE, ABC, ACD\}$ . Compute  $H_2$  (we know  $B_2$  and  $Z_2$  from last week). Let  $\sigma = CD + DE + CE$ , write out the elements of the equivalence class  $[\sigma]$ . Now add the 2-simplex  $CDE$  to form  $K'$ , without recomputation, determine  $H_1(K')$

- Sheryl

- Danya.